

Math 50

2.1 Expectation & variance 2.2 The Normal distribution

2.3 Linear regression: a conditionally Normal model 2.4 Coefficient of determination and correlation Problems

Unit 2

Expectation, the Normal distribution, and single-predictor linear regression model

In this unit we introduce **expectation**, an operation which takes a random variable and produces a deterministic quantity. The expectation of a random variable can be approximated with **sample averages**, and from them we can infer properties of the model (like its parameters). We then discuss the **Normal distribution** and a probability model built on the assumptions of conditional normality and linearity; namely, the **linear regression** model.

CONCEPTS

Expectation, variance and standard deviation, conditional expectation, empirical averages, coefficient of variation, linearity of expectation, the tower property and the law of total variance, the Normal distribution, standardization, linear transformations and sums of independent Normal random variables, the single-predictor linear regression model, exogenous vs. endogenous predictors, covariance, correlation, regression to the mean, and the coefficient of determination (R^2).

THINGS TO PRACTICE

- Calculate expectations by hand and translate between mathematical definitions and code (on exams you will not be asked to write code, but you may be asked to explain what a few lines of code do).
- Work with the Normal distribution: standardize a variable, compute probabilities using the bell curve shape (68/95/99.7), and find the distribution of a linear transformation or sum of independent Normal random variables.
- For a linear regression model with a binary predictor, understand why the regression coefficient is a difference of conditional averages.

- For a general predictor, relate the regression slope to covariance, and estimate it from data.
- State the assumptions of a linear regression model and identify (e.g. from plots) when they are violated.
- Recognize when a predictor may be endogenous, meaning the model's error term may still be related to the predictor.
- Use the correlation coefficient to explain why extreme values of a predictor go with less extreme predicted responses (regression to the mean).
- Compute and interpret the coefficient of determination (R^2) as the fraction of variance in Y explained by X , and relate it to correlation.

2.1 Expectation, variance and standard deviation

Sample averages and expectation

It's usually difficult to obtain the full distribution of a random variable from data, and it may not even be that relevant to the questions we're asking. Instead, we'd like to summarize properties of a random variable by looking at averages. In other cases, we have a good idea what type of distribution we're dealing with, but there are unknown parameters that can be estimated by averages.

You're probably already familiar with the **sample mean** (also called the **sample average** or **empirical average**). If $Y_1, Y_2, \dots, Y_n$ are iid samples of Y , the sample mean is

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

In code, if `x` is a NumPy array holding $Y_1, \dots, Y_n$, the sample mean is `np.mean(x)`.

More generally, we might look at the average of some function of a random variable,

$$\overline{g(Y)} = \frac{1}{n} \sum_{i=1}^n g(Y_i)$$

(a function of a random variable is just another random variable, so there's nothing too deep here). If we take

$$g(y) = 1_A(y) = \begin{cases} 1 & y \in A \\ 0 & y \notin A \end{cases}$$

then we connect the idea of a sample average to an estimate of a probability:

$$\overline{g(Y)} = \frac{N(y \in A)}{n} \approx P(\{Y \in A\}).$$

Any quantity we compute from data is, in some sense, a sample average — so a great deal of statistics is about understanding the behavior of sample averages.

Now we introduce the idea of **expectation**. It's helpful to think of expectation as the mathematical idealization of a sample average, just as probabilities are mathematical idealizations of long-run frequencies.

▼ Frequentist vs. Bayesian interpretation

Recall the flipped-but-hidden coin from [Unit 1](#). Whether you interpret its probability of heads as a long-run frequency or as your degree of belief (given the outcome is already fixed) is a key distinction in statistics. Thinking of probability as a long-run frequency is the frequentist interpretation and belief is Bayesian. We will say more about this in Units 6 and 7.

Suppose each Y_i is iid with sample space S . If n is large, the fraction of samples for which $Y_i = y$ will be $\approx P(\{Y = y\})$. We can express the sample average in terms of probabilities:

$$\begin{aligned} \bar{Y} &= \frac{1}{n} \sum_i Y_i = \frac{1}{n} \sum_{y \in S} y N(Y = y) \\ &= \sum_{y \in S} y \frac{N(Y = y)}{n} \approx \sum_{y \in S} y P(Y = y). \end{aligned}$$

The right-hand side is the definition of the mean, or expectation, denoted

$$E[Y] = \sum_y y P(Y = y).$$

To summarize what we just saw (which should be intuitively clear):

$$E[Y] \approx \bar{Y}.$$

Sometimes we write $\mathbb{E}$ instead of E to distinguish it from other variables named E . If we have a function $g : S \rightarrow S'$ from the sample space to some other space, then $g(Y)$ is a new random variable with sample space S' — but we don't usually need the distribution of $X = g(Y)$ to compute its expectation, since

$$E[X] = E[g(Y)] = \sum_{y \in S} g(y) P(Y = y).$$

This is convenient: we can use the distribution of Y to compute the expectation of X , which is often simpler (this will matter for the variance, below).

It's important to understand that, just like probabilities, expectation is an operation that takes a random variable to a deterministic number. The sample average is the approximate, empirical version of this — I like to think of expectations and sample averages as living in “math world” and “data world” respectively. In Unit 3 we'll make the connection between the two worlds precise via the law of large numbers and the CLT.

Example (expectation of a discrete random variable)

Let Y take values in $\{1, 2, 3\}$ with $P(Y = 1) = \frac{1}{2}$, $P(Y = 2) = \frac{1}{3}$, $P(Y = 3) = \frac{1}{6}$.

Question: what is $E[Y]$?

Solution:

$$E[Y] = \sum_{y=1}^3 y P(Y = y) = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{6} = \frac{1}{2} + \frac{2}{3} + \frac{1}{2} = \frac{5}{3}.$$

So $E[Y] = 5/3$.

The code below illustrates the sample mean converging to $5/3$ as n grows.

```

import numpy as np
import matplotlib.pyplot as plt

values = [1, 2, 3]
probs = [1/2, 1/3, 1/6]

Nmax = 5000
samples = np.random.choice(values, size=Nmax, p=probs)

sample_means = np.cumsum(samples) / np.arange(1, Nmax + 1)

plt.figure(figsize=(6,4))
plt.plot(sample_means, label="Sample mean")
plt.axhline(5/3, color="red", linestyle="--", label="Theoretical $E[Y]=5/3$")
plt.xlabel("n (sample size)")
plt.ylabel("Sample mean")
plt.legend()
plt.title("Convergence of sample mean to expectation")
plt.show()

```

This same convergence appears in the [Unit 1 Bernoulli simulation](#): the running sample frequency is just the sample mean of Bernoulli draws, so it settles near $E[Y] = q$.

Measuring variation

One of the most important expectations is the **variance**, which measures the typical squared distance from the mean:

$$\text{var}(Y) = E[(Y - E[Y])^2].$$

Another way to write this is

$$\text{var}(Y) = E[Y^2] - (E[Y])^2.$$

Example (mean and variance of a Bernoulli random variable)

Let $Y \sim \text{Bernoulli}(q)$, with $Y = 1$ with probability q .

Question: what are $E[Y]$ and $\text{var}(Y)$?

Solution:

$$E[Y] = P(Y = 0) \times 0 + P(Y = 1) \times 1 = q.$$

Similarly, you should be able to show $\text{var}(Y) = q(1 - q)$; you will also check this formula by simulation in a problem.

Based on this example, we can estimate q using $\hat{q} = \bar{Y}$, as expected — here $\hat{q}$ is shorthand for “an estimator of q .” (We will dive deeper into the concept of an estimator in Unit 3).

To measure “how much variation” there is in a random variable, we’d like to compare the variance to the mean. But there’s a problem: the variance has different units than the mean. If mean human height is about 170 cm, the variance might be around 100 cm^2 — hard to interpret, since it’s in squared centimeters. Taking the square root of the variance gives the **standard deviation**, which brings the spread back into the same units as the mean (here, 10 cm).

But even the standard deviation isn’t enough when comparing variability across different contexts. Suppose weight has mean 70 kg and standard deviation 10 kg. The “10” here isn’t directly comparable to the “10 cm” for height, because the scales are different — what matters is variation *relative to the mean*.

This leads to the **coefficient of variation (CV)**,

$$\text{CV} = \frac{\sigma}{\mu},$$

where σ is the standard deviation and μ is the mean. The CV is unitless, so it allows comparisons across variables measured in different units or with very different scales. For

example, a CV of 0.06 in height ($10/170$) indicates less relative variability than a CV of 0.14 in weight ($10/70$).

Quantity	Typical CV
Adult height	≈ 0.05
Adult weight	≈ 0.2
Household income	≈ 1
Household wealth	3–5

Conditional expectation

The **conditional expectation** is the expectation of the conditional distribution:

$$E[X \mid Y = y] = \sum_x x P(X = x \mid Y = y).$$

With samples $\{(x_1, y_1), \dots, (x_n, y_n)\}$ and at least one observation with $Y = y$,

$$E[X \mid Y = y] \approx \frac{1}{N(Y = y)} \sum_{i=1}^n 1_{\{y_i=y\}} x_i,$$

where $1_{\{y_i=y\}}$ is the **indicator function** (1 if $y_i = y$, 0 otherwise). Put simply: we compute a conditional expectation from data by taking the sample average among samples that satisfy the condition.

It should be clear that if X and Y are independent, then $E[X \mid Y = y] = E[X]$ for every y with $P(Y = y) > 0$ — but **the converse is false: it's possible for this to hold even when X and Y are not independent!**

Example ($E[X \mid Y] = E[X]$ without independence)

Let $Y \in \{-1, 1\}$ with $P(Y = 1) = P(Y = -1) = 1/2$. Given $Y = 1$, let $X \in \{-1, 1\}$ each with probability $1/2$; given $Y = -1$, let $X \in \{-2, 2\}$ each with probability $1/2$.

Question: compute $E[X | Y = 1]$ and $E[X | Y = -1]$. Are X and Y independent?

Solution:

$$E[X | Y = 1] = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0, \quad E[X | Y = -1] = \frac{1}{2}(-2) + \frac{1}{2}(2) = 0.$$

So $E[X | Y = y] = 0 = E[X]$ for both values of y – the conditional expectation doesn't depend on y at all. But X and Y are **not** independent: for instance $P(X = 1 | Y = 1) = \frac{1}{2}$ while $P(X = 1 | Y = -1) = 0$, since X can't even equal 1 when $Y = -1$. Matching conditional means only tells us the *average* of X doesn't shift with Y ; the whole *distribution* of X can still change with Y .

Example (computing a conditional expectation)

Consider (Y_A, Y_B) with the joint distribution from Unit 1:

$$P(Y_A = y_A, Y_B = y_B) = \begin{cases} 1/2 & y_A = 0, y_B = 0 \\ 1/8 & y_A = 0, y_B = 1 \\ 1/8 & y_A = 1, y_B = 0 \\ 1/4 & y_A = 1, y_B = 1 \end{cases}$$

Question: compute $E[Y_A | Y_B = 1]$.

Solution:

$$P(Y_A = 1 | Y_B = 1) = \frac{P(Y_A = 1, Y_B = 1)}{P(Y_B = 1)} = \frac{1/4}{3/8} = \frac{2}{3},$$

so $P(Y_A = 0 | Y_B = 1) = 1/3$ and $Y_A | (Y_B = 1) \sim \text{Bernoulli}(2/3)$, which means

$$E[Y_A | Y_B = 1] = \frac{2}{3}.$$

For the small dataset in the demo below,

$$\{(Y_1, Y_2)\} = \{(1, 2), (1, 2), (3, 1), (1, 4), (3, 3), (2, 2), (1, 5)\}.$$

To compute $E[Y_1 \mid Y_2 = 2]$, first keep only the rows with $Y_2 = 2$:

$$(1, 2), (1, 2), (2, 2).$$

Then average the Y_1 values in those rows:

$$E[Y_1 \mid Y_2 = 2] \approx \frac{1 + 1 + 2}{3} = \frac{4}{3}.$$

The demo repeats this same two-step procedure for a few different conditions: pick a conditional average, and it keeps the rows satisfying the condition and highlights the numbers being averaged. Those two steps are exactly the two steps in the formula above — restrict to the rows meeting the condition, then average the target column over them.

Conditional averages as filtered rows

row	Y_1	Y_2
1	1	2
2	1	2
3	3	1
4	1	4
5	3	3
6	2	2
7	1	5

$$E[Y_2 \mid Y_1 > 1] = (1 + 3 + 2)/3 = 2.00$$

Example (conditional expectation from data)

Consider data on children's test scores. The variable `kid_score` is the child's test score, and `mom_hs` is 1 if the mother graduated high school and 0 otherwise.

`pandas` is used only to load the CSV; the rest of the example works with the `numpy` arrays `kid_score` and `mom_hs`.

```
import numpy as np
import pandas as pd

url = (
    "https://raw.githubusercontent.com/"
    "avehtari/ROS-Examples/"
    "master/KidIQ/data/kidiq.csv"
)
df = pd.read_csv(url)
kid_score = df["kid_score"].to_numpy()
mom_hs = df["mom_hs"].to_numpy()
```

Let Y be the test score and let X be `mom_hs`. The conditional expectation $E[Y \mid X = 1]$ is the average test score among children whose mothers graduated high school. The conditional expectation $E[Y \mid X = 0]$ is the corresponding average among children whose mothers did not.

```
overall = np.mean(kid_score)
mean_given_hs0 = np.mean(kid_score[mom_hs == 0])
mean_given_hs1 = np.mean(kid_score[mom_hs == 1])

print("E[Y] estimated from all rows:", overall)
print("E[Y | X=0] estimated within group:", mean_given_hs0)
print("E[Y | X=1] estimated within group:", mean_given_hs1)
```

This code is doing the same operation as the definition: it restricts the rows to a condition, then averages Y inside that restricted sample space. If the two conditional

averages differ substantially, then $E[Y \mid X = x]$ depends on x , which is evidence that X and Y are not independent in this dataset.

Drill

Computing conditional averages

Suppose we have data on a pair (Y_1, Y_2) :

$$\{(1, 2), (1, 2), (3, 1), (1, 4), (3, 3), (2, 2), (1, 5)\}.$$

Compute each quantity by hand. Then describe, in one sentence for each part, exactly which rows are being averaged.

- $E[Y_1]$
- $E[Y_1 \mid Y_2 = 2]$
- $E[Y_2 \mid Y_1 = 1]$
- $E[Y_2 \mid Y_1 > 1]$

Conditional means from a joint table

Suppose (X, Y) has joint distribution

	$X = 0$	$X = 1$	$X = 2$
$Y = 0$	0.10	0.20	0.10
$Y = 1$	0.15	0.15	0.30

Compute $E[X]$, $E[X \mid Y = 0]$, and $E[X \mid Y = 1]$.

Properties of expectation

Expectation has several important properties, which become especially relevant once we work with linear regression models (defined in terms of conditional expectations).

1. **Linearity:** for random variables X and Y , $E[X + Y] = E[X] + E[Y]$.

▼ Proof

For discrete S_X, S_Y :

$$\begin{aligned} E[X + Y] &= \sum_{y \in S_Y} \sum_{x \in S_X} (x + y) P(X = x, Y = y) \\ &= \sum_x x \left(\sum_y P(X = x, Y = y) \right) + \sum_y y \left(\sum_x P(X = x, Y = y) \right) \\ &= E[X] + E[Y]. \end{aligned}$$

2. **Multiplication by a constant:** if a is a constant (not random), $E[aX] = aE[X]$.
3. **Factoring for independent variables:** if X and Y are independent, $E[XY] = E[X]E[Y]$.

▼ Proof

Using independence,

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy P(X = x, Y = y) = \sum_x \sum_y xP(X = x) yP(Y = y) \\ &= \left(\sum_x xP(X = x) \right) \left(\sum_y yP(Y = y) \right) = E[X]E[Y]. \end{aligned}$$

4. **Tower property:** for random variables X and Y , $E[E[X | Y]] = E[X]$, where $E[X | Y]$ is the random variable obtained by applying the deterministic function $f(y) = E[X | Y = y]$ to Y ; that is, $E[X | Y] = f(Y)$, so $E[E[X | Y]] = E[f(Y)]$.
5. **Law of total variance:** variance has an analogous decomposition,

$$\text{var}(X) = \underbrace{E[\text{var}(X | Y)]}_{\text{within-group variance}} + \underbrace{\text{var}(E[X | Y])}_{\text{between-group variance}}.$$

The total variance splits into the average spread *within* each group, plus how much the group *means* themselves vary.

Example (calculating conditional expectations with the tower property)

Let X indicate whether a student is in a morning section ($X = 0$) or afternoon section ($X = 1$). Suppose

$$P(X = 0) = 0.6, \quad P(X = 1) = 0.4,$$

and suppose quiz scores satisfy

$$E[Y | X = 0] = 78, \quad E[Y | X = 1] = 86.$$

Question: compute $E[Y]$ using the tower property, both by hand and with code.

Solution: by the tower property,

$$\begin{aligned} E[Y] &= E[E[Y | X]] \\ &= E[Y | X = 0]P(X = 0) + E[Y | X = 1]P(X = 1) \\ &= 78(0.6) + 86(0.4) = 81.2. \end{aligned}$$

The overall mean is a weighted average of the conditional means, with weights given by the group probabilities. The code below confirms the tower property numerically.

```

import numpy as np

rng = np.random.default_rng(2025)

N = 100_000
X = rng.choice([0, 1], size=N, p=[0.6, 0.4])
Y = np.empty(N)
Y[X == 0] = rng.normal(loc=78, scale=8, size=np.sum(X == 0))
Y[X == 1] = rng.normal(loc=86, scale=8, size=np.sum(X == 1))

EY_given_X0_hat = np.mean(Y[X == 0])
EY_given_X1_hat = np.mean(Y[X == 1])
pX0_hat = np.mean(X == 0)
pX1_hat = np.mean(X == 1)
EEY_given_X_hat = EY_given_X0_hat * pX0_hat + EY_given_X1_hat * pX1_hat

print(f"Theoretical E[Y]      = 81.200000")
print(f"Estimated E[Y]        = {np.mean(Y):.6f}")
print(f"Estimated E[E[Y|X]]     = {EEY_given_X_hat:.6f}")

```

Example (law of total variance)

Continuing the previous example, suppose additionally that quiz scores have the same spread in each section, $\text{var}(Y \mid X = 0) = \text{var}(Y \mid X = 1) = 64$.

Question: compute $\text{var}(Y)$ using the law of total variance.

Solution: the within-group term is the weighted average of the (here, equal) conditional variances:

$$E[\text{var}(Y \mid X)] = 0.6(64) + 0.4(64) = 64.$$

The between-group term is the variance of the conditional means 78 and 86 around the overall mean $E[Y] = 81.2$, weighted by $P(X = 0) = 0.6$ and $P(X = 1) = 0.4$:

$$\text{var}(E[Y \mid X]) = 0.6(78 - 81.2)^2 + 0.4(86 - 81.2)^2 = 6.144 + 9.216 = 15.36.$$

So

$$\text{var}(Y) = 64 + 15.36 = 79.36.$$

Even though every section has the *same* internal spread, the overall variance is bigger than either conditional variance, because the two sections also have different means. The code below extends the previous simulation to confirm this.

```
within_hat = pX0_hat * np.var(Y[X == 0]) + pX1_hat * np.var(Y[X == 1])
between_hat = (
    pX0_hat * (EY_given_X0_hat - np.mean(Y)) ** 2
    + pX1_hat * (EY_given_X1_hat - np.mean(Y)) ** 2
)

print(f"Theoretical var(Y)                = 79.360000")
print(f"Estimated var(Y)                  = {np.var(Y):.6f}")
print(f"Estimated E[var(Y|X)] + var(E[Y|X]) = {within_hat + between_hat:.6f}")
```

Example (expectation of a sum)

Let $Y = \sum_{i=1}^N X_i$ with $X_i \sim \text{Bernoulli}(q)$ iid. If you read the optional section in Unit 1, you will recognize that this means $Y \sim \text{Binomial}(N, q)$.

Question: what are $E[Y]$ and $\text{var}(Y)$?

Solution: by linearity (property 1),

$$E[Y] = E\left[\sum_{i=1}^N X_i\right] \stackrel{(1)}{=} \sum_{i=1}^N E[X_i] = Nq.$$

Similarly,

$$E[Y^2] = E\left[\left(\sum_i X_i\right)^2\right] = E\left[\sum_i \sum_j X_i X_j\right] \stackrel{(1)}{=} \sum_i \sum_j E[X_i X_j]$$

$$\stackrel{(3)}{=} \sum_i \sum_{j \neq i} q^2 + \sum_i E[X_i^2] = N(N-1)q^2 + Nq,$$

using $E[X_i^2] = E[X_i] = q$ and property 3 (factoring) for the $i \neq j$ terms. Therefore

$$\text{var}(Y) = E[Y^2] - E[Y]^2 = N(N-1)q^2 + Nq - N^2q^2 = Nq(1-q).$$

To summarize:

$$E[Y] = qN, \quad \text{var}(Y) = Nq(1-q).$$

Both the mean and variance grow linearly with N , but the standard deviation only grows like $\sqrt{N}$. Relative to the mean, the spread shrinks, which is captured by the CV:

$$\text{CV} = \frac{\sqrt{\text{var}(Y)}}{E[Y]} = \sqrt{\frac{1-q}{q} \cdot \frac{1}{N}}.$$

This measures variation *relative* to the average, which matters for many applications.

Example (election modeling)

Consider an election between two candidates. Let q be the population fraction supporting candidate one, and suppose N people vote (with N much smaller than the population, since turnout is low). Each voter's ballot is a $\text{Bernoulli}(q)$ indicator of support for candidate one, so the number of votes for candidate one, M , is a sum of N iid $\text{Bernoulli}(q)$ draws, i.e. $M \sim \text{Binomial}(N, q)$, with mean and variance as derived above.

Question: in a city where $q = 0.51$ support a candidate, and $N = 1000$ vote, what's the chance the vote share $\hat{q} = M/N$ differs from q by more than 1%? There's no closed form for this, so we answer by simulation.

```
import numpy as np

q = 0.51
N = 1000
delta = 0.01
trials = 200_000

rng = np.random.default_rng(123)

M_samples = rng.binomial(n=N, p=q, size=trials)
q_hat = M_samples / N
prob_est = np.mean(np.abs(q_hat - q) > delta)

print(f"Monte Carlo estimate P(|q_hat - {q}| > {delta}) = {prob_est:.6f}")
```

In this problem, the vote share is exactly the sample mean of the individual votes, $\hat{q} = M/N$, so $E[\hat{q}] = q$. The variance is

$$\text{var}(\hat{q}) = \text{var}(M/N) = \frac{1}{N^2} \text{var}(M) = \frac{q(1-q)}{N},$$

which tends to zero as $N \rightarrow \infty$, while $E[\hat{q}]$ doesn't depend on N at all. This is a consequence of the CV decreasing with N , and it's what lets us determine q by approximating $E[\hat{q}]$ with the sample mean.

Drill

Expectation and variance from a table

Let Y have distribution

$$P(Y = 0) = 0.2, \quad P(Y = 1) = 0.5, \quad P(Y = 2) = 0.3.$$

Compute $E[Y]$, $E[Y^2]$, and $\text{var}(Y)$.

Weighted conditional means

Suppose $P(X = 0) = 0.7$, $P(X = 1) = 0.3$, $E[Y | X = 0] = 10$, and $E[Y | X = 1] = 20$. Use the tower property to compute $E[Y]$.

Law of total variance from conditional moments

Suppose $P(X = 0) = 0.7$, $P(X = 1) = 0.3$, $E[Y | X = 0] = 10$, $E[Y | X = 1] = 20$, $\text{var}(Y | X = 0) = 4$, and $\text{var}(Y | X = 1) = 9$. Use the law of total variance to compute $\text{var}(Y)$.

Sum of Bernoulli variables

Let $Y = \sum_{i=1}^{12} X_i$, where the X_i are iid Bernoulli(0.25). Compute $E[Y]$, $\text{var}(Y)$, and the coefficient of variation.

2.2 The Normal distribution

We now define the most important distribution in statistics: the **Normal distribution**.

Recall from [Unit 1](#) that a continuous random variable Y is characterized by a *density* $f(y)$, with $P(a < Y < b) = \int_a^b f(y) dy$. The Normal distribution is continuous, symmetric, bell-shaped, and completely determined by two parameters: its mean μ and variance σ^2 . Its density is

$$g(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

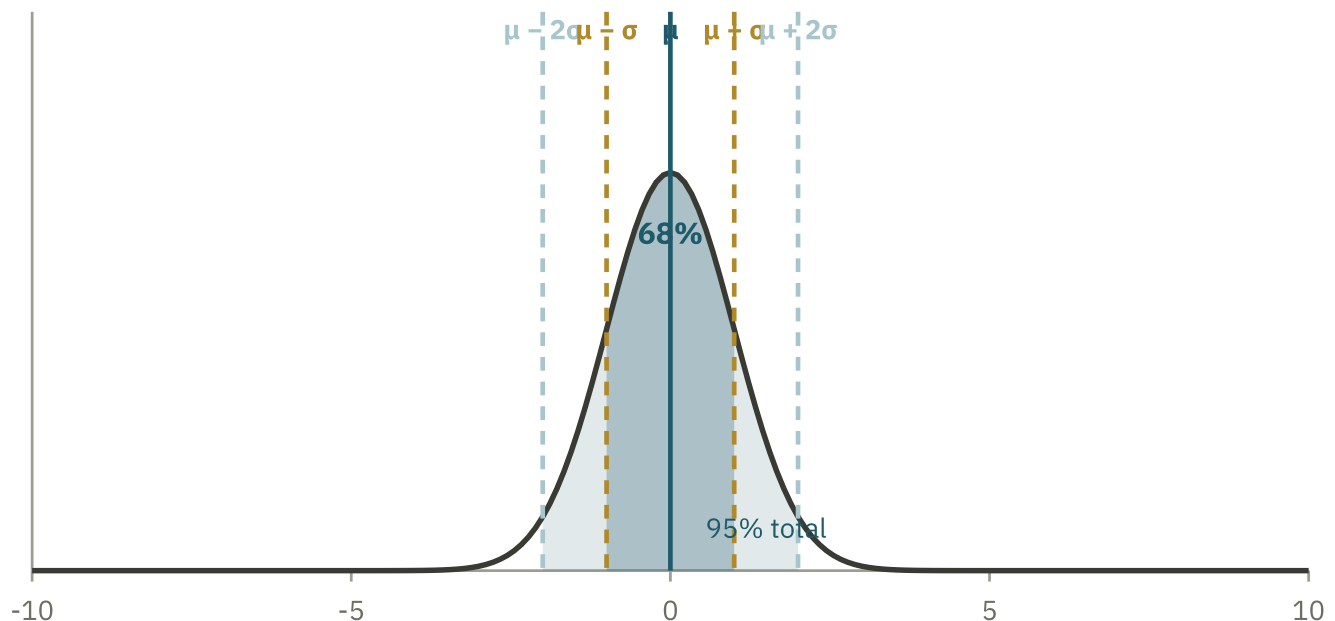
Despite the simplicity of this formula, calculating Normal probabilities by hand (integrating to find area under the curve) is difficult. Instead we rely on rough rules of thumb: about 68% of the probability lies within 1 standard deviation of the mean, about 95% lies within 2 standard deviations, and about 99.7% lies within 3 standard deviations. The demo below keeps the x -axis fixed while you change μ and σ , so you can see that μ shifts the center

and σ changes the spread. Its two shaded bands are $\mu \pm \sigma$ and $\mu \pm 2\sigma$, and they hold about 68% and 95% of the area wherever you put the sliders — a direct consequence of the Normal distribution's bell curve shape.

If X has density $g(x)$ above, we write $X \sim \text{Normal}(\mu, \sigma^2)$. It can be shown that $E[X] = \mu$ and $\text{var}(X) = \sigma^2$ — hence the name. **Be careful:** sometimes (in code or in math) Normal random variables are parameterized by mean and standard deviation instead of mean and variance, so you'll also see $\text{Normal}(\mu, \sigma)$. Both conventions are used, so always check.

In code, `np.random.normal(mu, sigma)` (or, with a `Generator`, `rng.normal(mu, sigma)`) draws a sample from $\text{Normal}(\mu, \sigma^2)$ — note it takes the standard deviation `sigma`, not the variance.

Normal distribution: mean and standard deviations



Example (calculating Normal probabilities)

Suppose $Y \sim \text{Normal}(5, 4)$.

Question: what is (approximately) $P(Y > 7)$?

Solution: since $5 + 2 = 7$, this is asking how likely a Normal variable is to exceed 1 standard deviation above the mean — about 16%. (We can always compute this exactly in Python too.)

Question: what is $P(Y > 3 \mid Y < 7)$?

Solution:

$$P(Y > 3 \mid Y < 7) = \frac{P(Y > 3, Y < 7)}{P(Y < 7)}.$$

Since $3 = 5 - 2 = \mu - \sigma$ and $7 = 5 + 2 = \mu + \sigma$, we have $P(Y < 7) \approx 0.841$ and $P(Y > 3, Y < 7) \approx 0.682$, so the answer is about 0.81.

Properties of Normal random variables

Linear transformations of Normal random variables. Suppose $Z \sim \text{Normal}(0, 1)$ and define $X = \sigma Z + \mu$. Write $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$ for the density of Z — this is the density g above with $\mu = 0, \sigma = 1$. Then

$$P(X < x) = P(Z < \frac{x-\mu}{\sigma}) = \int_{-\infty}^{(x-\mu)/\sigma} \phi(z) dz.$$

Substituting $u = \mu + \sigma z$ (so $dz = du/\sigma$), when $z \rightarrow -\infty$ we get $u \rightarrow -\infty$, and at $z = (x - \mu)/\sigma$ we get $u = x$:

$$P(X < x) = \int_{-\infty}^x \phi\left(\frac{u-\mu}{\sigma}\right) \frac{1}{\sigma} du = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(u-\mu)^2}{2\sigma^2}\right) du.$$

This shows $X \sim \text{Normal}(\mu, \sigma^2)$ — in particular, any Normal random variable is a linear transformation of a standard normal. More generally, if $X \sim \text{Normal}(\mu_x, \sigma_x^2)$ and $Y = aX + b$, then Y should also be Normal. Taking expectations, $E[Y] = a\mu_x + b$, and since $\text{var}(aX) = a^2 \text{var}(X)$ and $\text{var}(b) = 0$,

$$Y \sim \text{Normal}(a\mu_x + b, a^2\sigma_x^2).$$

Going from Z to X to Y , we're just multiplying and shifting — think about what that does to a histogram. The process of going from X to Z is called **standardizing**: for any X , the standardized variable is

$$Z = \frac{X - \mu_x}{\sigma_x}.$$

Transforming X to a standard normal is equivalent to measuring X in units of standard deviations — if you make a histogram of X , this transformation just relabels the x -axis in those units.

Theorem (sums of independent Normal random variables). If $X_1 \sim \text{Normal}(\mu_1, \sigma_1^2)$ and $X_2 \sim \text{Normal}(\mu_2, \sigma_2^2)$ are independent, then

$$aX_1 + bX_2 + d \sim \text{Normal}(a\mu_1 + b\mu_2 + d, a^2\sigma_1^2 + b^2\sigma_2^2).$$

We'll use this in Unit 3, where via the Central Limit Theorem, we can approximate the distribution of almost any sample average with a Normal distribution.

Drill

Normal probabilities

Suppose $Y \sim \text{Normal}(10, 9)$.

- Write the standardized version of Y .
- Estimate $P(Y > 13)$ using the bell curve shape.
- Estimate $P(7 < Y < 13)$ using the bell curve shape.

Linear transformation

Suppose $X \sim \text{Normal}(5, 9)$ and define $Y = -2X + 7$.

- Find the distribution of Y by giving its mean and variance.
- Standardize Y , then use the bell curve shape to estimate $P(Y > 3)$.

- c. Without computing anything, write down $P(Y > -3)$ and explain in one sentence how you know.

Sum of Normal random variables

Suppose $X_1 \sim \text{Normal}(2, 4)$ and $X_2 \sim \text{Normal}(-1, 9)$ are independent, and define $S = 3X_1 - 2X_2$.

- Find the distribution of S by giving its mean and variance.
- Standardize S .
- Use the bell curve shape to estimate $P(0 < S < 16)$.

2.3 Linear regression: a conditionally Normal model

Equipped with conditional expectation and the Normal distribution, we can define our first regression model. A (single-predictor) **linear regression model** is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2),$$

where X is the **predictor** and Y is the **response variable**.

The conditional expectation is the line

$$E[Y \mid X] = \beta_0 + \beta_1 X,$$

and the Normal distribution describes the vertical spread of Y around that line. Equivalently, we can write

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma^2),$$

where ϵ is the noise or error term. Let us summarize the assumptions being made in this model:

- Linearity:** the conditional mean $E[Y \mid X] = \beta_0 + \beta_1 X$ is a straight line in X .

- b. **Homoskedasticity:** the conditional variance $\text{var}(Y | X) = \sigma^2$ is the same at every value of X .
- c. **Conditional normality:** $Y | X$ (equivalently ϵ) is Normally distributed.
- d. **Exogeneity of ϵ :** $E[\epsilon | X] = 0$, so after accounting for X , the leftover error has mean zero at every value of X . A predictor satisfying this is called **exogenous**; if it fails, X is **endogenous**.

For example, suppose X indicates whether a student attended a study-skills workshop and Y is their exam score. If X is assigned by a coin flip (a randomized trial), the unmeasured factors drive ϵ are equally split between the two groups. Therefore $E[\epsilon | X] = 0$. But if students instead self-select into the workshop, the more motivated or more anxious students may be the ones who choose to attend and X endogenous.

Example (binary predictor: the difference of means)

Let $X \sim \text{Bernoulli}(1/2)$ and $Y | X \sim \text{Normal}(\beta_1 X + \beta_0, \sigma^2)$ — motivated by a clinical trial where participants are placed in control ($X = 0$) or treatment ($X = 1$) groups, and Y is some measured outcome (e.g. blood pressure).

Writing the conditional expectation at the two possible values of X gives

$$E[Y | X = 0] = \beta_0, \quad E[Y | X = 1] = \beta_0 + \beta_1.$$

Therefore

$$E[Y | X = 1] - E[Y | X = 0] = \beta_1.$$

For a binary predictor, the slope β_1 is the difference between the treatment-group mean and the control-group mean. In data, the natural estimate is the same comparison with conditional sample averages:

$$\hat{\beta}_1 = \overline{Y | X = 1} - \overline{Y | X = 0},$$

$$\overline{Y | X = 1} = \frac{1}{N(X = 1)} \sum_{i=1}^n Y_i 1_{X_i=1},$$

where n is the number of samples and $N(X = 1)$ counts those with $X_i = 1$.

The functions below implement these expressions in code.

```
import numpy as np

def generate_data(beta0, beta1, sigma, n_samples):
    x = np.random.choice([0, 1], n_samples)
    y = beta0 + beta1 * x + np.random.normal(0, sigma, n_samples)
    return x, y

def beta1_hat(x, y):
    return y[x == 1].mean() - y[x == 0].mean()
```

Example (linear regression with a Normal predictor)

Let $X \sim \text{Normal}(\mu_x, \sigma_x^2)$ and $Y \mid X \sim \text{Normal}(\beta_1 X + \beta_0, \sigma^2)$. Now we can't estimate β_1 by comparing means at $X = 0$ vs. $X = 1$, since we may never observe those exact values — we'll need another approach, leading to least squares.

Question: what's the marginal distribution of Y ? What is $E[XY]$, and how does it compare to $E[X]E[Y]$?

Solution: write $Y = \beta_1 X + \beta_0 + Z$ with $Z \sim \text{Normal}(0, \sigma^2)$ independent of X . Then the marginal distribution of Y is a sum of two independent Normals, so

$$Y \sim \text{Normal}(\beta_1 \mu_x + \beta_0, \beta_1^2 \sigma_x^2 + \sigma^2).$$

For $E[XY]$: since $E[XY \mid X = x] = xE[Y \mid X = x]$,

$$E[XY] = E[X E[Y \mid X]] = E[X(\beta_1 X + \beta_0)] = \beta_1 E[X^2] + \beta_0 E[X].$$

Using $E[X^2] = \text{var}(X) + E[X]^2 = \sigma_x^2 + \mu_x^2$,

$$E[XY] = \beta_1 \sigma_x^2 + \beta_1 \mu_x^2 + \beta_0 \mu_x.$$

Meanwhile $E[X]E[Y] = \mu_x(\beta_1\mu_x + \beta_0) = \beta_1\mu_x^2 + \beta_0\mu_x$. The difference between the two is the extra term $\beta_1\sigma_x^2$, which comes from the variance of X .

Covariance and estimating the slope

$$E[XY] - E[X]E[Y]$$

The quantity  from the example above is important enough to have its own name, the **covariance**,

$$\text{cov}(X, Y) := E[XY] - E[X]E[Y].$$

(Replacing Y with X recovers the variance: $\text{cov}(X, X) = \text{var}(X)$.) The example above shows that for *any* linear regression model $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2)$, regardless of the distribution of X (as long as $\text{var}(X) = \sigma_X^2 < \infty$),

$$\text{cov}(X, Y) = \beta_1 \sigma_X^2.$$

This is useful because it lets us estimate β_1 from data $(x_1, y_1), \dots, (x_n, y_n)$ without needing X to be binary:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

In Python, `np.cov(x, y)[0, 1]` computes the sample covariance and `np.cov(x, y)[0, 0]` the sample variance of `x`, so `np.cov(x, y)[0, 1] / np.cov(x, y)[0, 0]` estimates β_1 directly. (We'll return to this formula in Unit 3, where it's called the *least squares* estimator.)

Regression to the mean

Standardizing both variables, $Z_X = (X - \mu_X)/\sigma_X$ and $Z_Y = (Y - \mu_Y)/\sigma_Y$, turns the slope into a unitless quantity

$$\rho := \frac{\beta_1 \sigma_X}{\sigma_Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y},$$

called the **correlation** between X and Y (we'll study its properties, including why $-1 \leq \rho \leq 1$, in Unit 3). A short calculation shows $E[Z_Y | Z_X] = \rho Z_X$.

Whenever $|\rho| < 1$, this means an unusually large Z_X predicts a *less* extreme Z_Y — shrunk toward 0 by the factor ρ . This is **regression to the mean**: extreme values of X tend to go with less extreme values of Y , on average.

Example (regression to the mean)

Suppose exam 1 and exam 2 scores, standardized, have correlation $\rho = 0.6$.

Question: a student scores 2 standard deviations above the mean on exam 1. What's their predicted standardized score on exam 2?

Solution: $E[Z_Y | Z_X = 2] = 0.6 \times 2 = 1.2$. The predicted score is still above average, but less extreme than exam 1's — not because the student got worse, but because an unusually high score is partly luck, and luck doesn't repeat.

Drill

Binary predictor means

Suppose $X \in \{0, 1\}$ and

$$Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2).$$

Compute $E[Y | X = 0]$, $E[Y | X = 1]$, and $E[Y | X = 1] - E[Y | X = 0]$. Then explain why the difference of conditional sample averages is a natural estimator of β_1 .

Covariance, correlation, and shrinkage

Suppose $\text{var}(X) = 4$, $\text{var}(Y) = 25$, and $\text{cov}(X, Y) = 8$.

a. Compute β_1 .

- b. Compute the correlation ρ .
- c. If X is 2 standard deviations above its mean, what is the predicted standardized value of Y ?

2.4 Coefficient of determination and correlation

Coefficient of determination

In most applications the goal of regression modeling is to predict Y from X , so it's natural to look for a metric of how well these predictions can be made. Think about the best-case scenario: $\sigma_\epsilon = 0$ (and $\beta_1 \neq 0$), so Y is *deterministic* given X . Here σ_ϵ is the same noise standard deviation σ from the model definition above, but we want to distinguish between σ_X and σ_Y .

Predictions get worse as σ_ϵ grows, since σ_ϵ measures the variability in Y for a *fixed* X . However large relative to what? σ_ϵ^2 has the units of Y *squared*, so we should compare it to something with those same units: the marginal variance $\sigma_Y^2 = \text{var}(Y)$. Note (check this yourself!) that

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \geq \sigma_\epsilon^2,$$

i.e. the total variance in Y splits into variation from ϵ (factors unrelated to X) and $\beta_1^2 \sigma_X^2$ (the spread in X , weighted by the squared slope). This is actually a special case of the law of total variance from [Section 2.1](#). (Check this as well!)

This motivates the **coefficient of determination**,

$$\rho^2 = 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2} = 1 - \frac{\text{var}(Y | X)}{\text{var}(Y)},$$

which lies between 0 and 1. When $\rho^2 = 1$, essentially all the variation in Y comes from variation in X . When $\rho^2 = 0$ (i.e. $\sigma_\epsilon^2 = \sigma_Y^2$), knowing X doesn't change how variable Y is — all the variation in Y is due to something other than X .

The sample estimate of ρ^2 is denoted R^2 :

$$R^2 = 1 - \frac{\sum_{i=1}^n r_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2},$$

which should make sense as the natural estimator, given the definition above.

Correlation

[Section 2.3](#) already introduced the squared correlation coefficient

$\rho = \beta_1 \sigma_X / \sigma_Y = \text{cov}(X, Y) / (\sigma_X \sigma_Y)$ to explain regression to the mean. Here we show how to view this as the square of a signed quantity (positive or negative). Standardize the predictor and response,

$$Z_X = \frac{X - \mu_X}{\sigma_X}, \quad Z_Y = \frac{Y - \mu_Y}{\sigma_Y}.$$

This is natural since we want a dimensionless measure of association. Since

$$\mu_Y = \beta_0 + \beta_1 \mu_X,$$

$$Z_Y = \frac{\beta_0 + \beta_1 X + \epsilon - \mu_Y}{\sigma_Y} = \frac{\beta_1 (X - \mu_X)}{\sigma_Y} + \frac{\epsilon}{\sigma_Y} = \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y},$$

so

$$Z_Y \mid Z_X \sim \text{Normal}\left(\frac{\beta_1 \sigma_X}{\sigma_Y} Z_X, \frac{\sigma_\epsilon^2}{\sigma_Y^2}\right).$$

This is a simple regression of Z_Y on Z_X with slope $b = \beta_1 \sigma_X / \sigma_Y$, representing the expected change (in standard deviations of Y) associated with a one-standard-deviation change in X .

This motivates defining the **correlation coefficient** ρ as this slope:

$$\rho := b = \frac{\beta_1 \sigma_X}{\sigma_Y}.$$

Using $\text{cov}(X, Y) = \beta_1 \sigma_X^2$ from [Section 2.3](#),